1. Introduction

The purpose of this project is to perform **Exploratory Data Analysis (EDA)** on the Student Performance dataset. EDA helps us understand the structure of the dataset, identify missing values and outliers, study the relationships between variables, and gain useful insights before applying machine learning models.

2. Objective

The objectives of this EDA are:

- To understand the dataset.
- To check the size and structure of the data.
- To identify missing values and duplicate records.
- To detect and handle outliers.
- To calculate basic statistical measures.
- To visualize the data using different graphs.
- To understand the relationship between student characteristics and their performance.

3. Dataset Overview

The dataset was loaded using the **Pandas** library.

The following operations were performed:

- Displayed the first five rows using `head()`.
- Displayed the last five rows using `tail()`.
- Checked the dataset dimensions using `shape`.
- Displayed column names using `columns`.
- Examined data types using `info()`.

These steps helped in understanding the dataset before analysis.

4. Data Cleaning

Missing Values

The dataset was checked using:

PYTHON:

```
st.isnull().sum()
```

Observation:

- No missing values were found in the dataset.

Duplicate Records

Duplicate rows were checked using:

PYTHON:

```
st.duplicated().sum()
```

Duplicate records were removed using:

PYTHON:

```
st.drop_duplicates(inplace=True)
```

Observation:

- Duplicate records were removed to improve data quality.

5. Outlier Detection

Outliers in the **G3 (Final Grade)** column were detected using the **Interquartile Range (IQR)** method.

The following steps were performed:

- Calculated Q1 and Q3
- Calculated IQR
- Computed lower and upper limits
- Identified outliers
- Removed outlier rows
- Applied clipping to keep values within the acceptable range

Observation:

- Outliers were identified in the final grade column.
- Removing extreme values makes the dataset more reliable for analysis.

6. Statistical Analysis

The following statistical measures were calculated for the **Age** column.

Mean

The average age of students was calculated.

Median

The middle value of student age was calculated.

Mode

The most frequently occurring age was identified.

Variance

Variance measured the spread of student ages.

Standard Deviation

Standard deviation showed how much the ages vary from the average age.

7. Correlation Analysis

The notebook attempted to analyze the relationship between **Age** and **Final Grade (G3)**.

PYTHON:

np.correlate(st["age"], st["G3"])

Observation:
This gives a numerical comparison between Age and Final Grade. For stronger EDA, a correlation matrix (using `corr()`) would generally provide a clearer interpretation.

8. Data Visualization

Several visualizations were created to understand the dataset.

Bar Chart

A bar chart was plotted to display the number of male and female students.

Observation:

- The graph shows the gender distribution of students.

Line Chart

A line chart was created for the gender count.

Observation:

- It shows the variation in category counts, although a line chart is not the best choice for categorical data.

Histogram

A histogram was generated.

Observation:

- Histograms are mainly used for continuous numerical data. Using them for gender counts is not very informative.

Scatter Plot

A scatter plot was drawn between **G1** and **G2**.

Observation:

- Students who scored high in G1 generally also scored high in G2.
- This indicates a positive relationship between the two grades.

9. Key Findings

- The dataset was successfully loaded and explored.
- No missing values were found.
- Duplicate records were removed.
- Outliers in the final grade were detected and handled using the IQR method.
- Basic statistical measures were calculated for student age.
- Gender distribution was visualized using bar, line, and histogram plots.
- A scatter plot showed a positive relationship between G1 and G2 marks.
- The dataset became cleaner and more suitable for further machine learning tasks after preprocessing.